Digital Twin for Type 2 Diabetes

Forecasting high glucose two hours ahead by fusing health records with wearable data

Tailwind | IIM BODHGAYA | Happiest Health Digital Twin Challenge 2026

The problem and the use case

- Type 2 diabetes is a major chronic burden in India; care is reactive, with glucose spikes noticed after the fact
- Use case: for a patient whose glucose is in range now, will it exceed 180 mg/dL within 2 hours?
- A digital twin combines two streams: the health record (static) and wearable data (live)
- Goal: a working model that ingests both streams, warns early, and a doctor-facing dashboard
- Privacy: no real patient data (DPDP Act, HIPAA); records from Synthea, wearables simulated

How the data was built

- Synthea: 10,000 patients generated, 1,015 living type 2 diabetes patients kept (706 train, 155 validation, 150 test, split by patient)
- Each record sets hidden physiology (insulin sensitivity, beta-cell function); the record explains only part of it, so sensors add information
- Indian meal profiles, steps, sleep and stress feed a glucose-insulin simulator, calibrated to each patient's HbA1c
- Sensors: CGM with noise and dropouts, heart rate, HRV, steps, sleep stages
- Events are read off the simulated glucose, never injected; 603,022 labelled moments

Model and results (150 unseen test patients)

Model	AUROC
Naive rule: glucose now + trend	0.639
Health record only	0.675
Heart rate, HRV, steps, sleep only	0.739
CGM only	0.844
Fused: CGM + wearables + record	0.844

- Fused model: gradient-boosted trees on sensor-history features plus the record
- Probabilities are well calibrated (ECE under 0.01)
- Each version was checked on validation data first; ablations tested by patient-level bootstrap

Where the health record helps, and what an alert costs

EHR gain in AUROC

Situation	Gain
Full CGM history	about 0
Only 2 hours of CGM	+0.02
No CGM, cheap wearable only	+0.07 to +0.08

Alert settings (continuous monitoring)

Setting	Alerts/day	Warned 30+ min
Strict	1.7	26%
Balanced	2.6	39%
Sensitive	3.5	53%

- Record matters most when sensor data is limited, which fits India where CGM is not universal
- Median warning lead time 70 minutes; 71% of episodes get some warning at the balanced setting
- One global threshold under-alerts insulin users; calibrating to each patient's first week restores their warnings

The doctor dashboard

- Care team view: all patients sorted by urgency with live status
- Patient monitor: glucose, 2-hour risk, heart rate, HRV, steps and sleep, with alerts marked and live replay
- Explanation panel: which signal groups push the forecast up or down
- What-if lab: add a meal, add a walk or skip a meal and compare the simulated glucose with the usual course
- Adjustable alert sensitivity and threshold policy; weekly summary export

Limitations and next steps

- All data is synthetic: results show how learnable the simulated dynamics are, not clinical validity
- Synthea's diabetic patients are mostly well controlled, unlike many Indian cohorts
- What-if twin uses simulator parameters instead of fitting them from CGM
- Time of day still matters: CGM-only AUROC 0.844 with the clock, 0.731 without
- Next: external check on real open CGM data, fit the twin per patient, optional meal logging, clinician testing